Investment Memo Template (Dataset-Agnostic)

QM 2023 Capstone Project - Milestone 4

MEMORANDUM

TO: [Your Decision-Maker Audience - Choose One:]

- **Investment Committee** (REITs, stocks, crypto): Portfolio allocation decisions
- **Policy Committee** (economic indicators, labor data): Policy recommendations
- **Risk Committee** (banking, insurance data): Risk management guidance
- **Executive Team** (firm-level data): Strategic business decisions
- **Research Community** (academic framing): Contributions to literature

FROM: [Team Name: List all team member names] **DATE:** [Submission Date] **RE:** [Your Research Question + Recommendation Context]

REIT Example: *REIT Return Sensitivity to Interest Rate Policy – Investment Strategy Recommendation* **Crypto Example:** *Cryptocurrency Regulatory Risk Assessment – Portfolio Risk Management* **Housing Example:** *Housing Market Response to Monetary Policy – Policy Briefing for Fed Analysts*

Executive Summary

[2-3 paragraphs, approximately 0.5 page. No tables or figures. Use business language appropriate for your audience.]

Structure Template

[Paragraph 1: State the key finding in plain language]

- What did you analyze? ([N] entities, [time period])
- What did you find? (e.g., "A [X unit] increase in [DRIVER] changes [OUTCOME] by [Y units]")
- What's the causal mechanism? (List 2-3 economic channels)

[Paragraph 2: Group-level insights OR robustness evidence]

- **If your dataset has groups** (sectors, regions, asset types): Which groups are most/least sensitive?
- **If no natural grouping**: What robustness checks confirm the main finding?
- Magnitude of differential effects or evidence of stability

[Paragraph 3: Recommendation for decision-maker]

- Given current environment, what should decision-makers do?
- Specific, actionable advice (not vague "be cautious")
- Forward-looking guidance (scenario analysis)

Example: REIT Investment Committee

> Using a panel of 532 US equity REITs over 2015-2023, we find that a 1 percentage point increase in the Federal Funds Rate reduces REIT returns by 2.5 percentage points ($p < 0.01$), controlling for REIT and time fixed effects. This effect occurs with a 2-month lag and is driven by leverage costs and discount rate increases. Sector analysis reveals that Retail and Office REITs are highly sensitive (correlations: -0.45, -0.42), while Industrial REITs are resilient (-0.15).
>
> Given the current elevated rate environment (5.0-5.25%), we recommend a **15% overweight** in Industrial REITs and a **neutral-to-underweight** stance in Retail and Office REITs. If the Federal Reserve cuts rates by 100 basis points over the next 12 months, our model predicts a 2.5% recovery in aggregate REIT returns.

Example: Cryptocurrency Risk Committee

> Analyzing 150 cryptocurrencies over 2020-2024, we find that negative regulatory announcements (SEC enforcement actions, exchange restrictions) reduce token returns by 8.3 percentage points within one week ($p < 0.001$), controlling for token and time fixed effects. This effect is driven by liquidity channel disruptions (reduced exchange listings) and institutional investor exit. DeFi tokens exhibit higher sensitivity (-12.1%) than centralized exchange tokens (-5.4%), reflecting differential regulatory risk exposure.
>
> Given ongoing regulatory uncertainty (SEC classification debates, stablecoin regulation), we recommend **reducing exposure to DeFi tokens by 20%** and **increasing allocation to Bitcoin/Ethereum** (lower regulatory risk due to established compliance frameworks). If Congress passes comprehensive crypto legislation (30% probability over 12 months), our model predicts a 15-20% recovery in high-risk DeFi assets.

Using This Template with AI Assistance

This template uses REIT examples because that's the course default dataset. If using alternative data (crypto, housing, macro, firm-level), follow this translation workflow:

Step 1: Identify Conceptual Equivalent

AI Prompt Template:

```
I'm working on a Milestone 4 Investment Memo for a panel data course. The REIT
example uses
'[REIT_VARIABLE]' which measures [CONCEPT]. My dataset is [DATASET_NAME] at
[ENTITY-TIME LEVEL].
What variable in my dataset measures the same concept? List 2-3 candidates and
explain the
economic mechanism.
```

Verification Checklist:

- ☐ I understand why each suggested variable measures this concept
- ☐ I checked variable distribution (`df[X].describe()`) to ensure units make sense

- ☐ I found 1+ academic paper using this variable in my domain (cite in References)
- ☐ I can explain this to a classmate unfamiliar with my dataset

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Step 2: Translate Economic Interpretation

AI Prompt Template:

```
The REIT memo says: '[REIT_INTERPRETATION]'. Translate this to my dataset
([DATASET_NAME])
using variable [MY_VAR]. Ensure units are correct and the economic mechanism is
domain-appropriate.
```

Verification Checklist:

- ☐ My interpretation uses domain-appropriate terminology (not REIT jargon)
- ☐ Units are explicit and match my data (percentage points, basis points, dollars, etc.)
- ☐ I can explain the causal mechanism without looking at AI output
- ☐ I checked the magnitude against domain literature (e.g., crypto returns are more volatile than REITs)

Step 3: Document in AI Audit

- Record all translation prompts and AI responses
- Explain what you chose vs. what AI suggested
- **Critique:** What did AI miss? What domain knowledge did you add?

Common Pitfall: Blind find-replace of variable names without understanding mechanisms. Example:

- ❌ Bad: "A 1% increase in Bitcoin price reduces Ethereum returns..." (nonsensical)
- ☑ Good: "A 1 percentage point increase in regulatory severity reduces DeFi token returns..." (causal mechanism)

Common Translations Reference

REIT Concept	Crypto Equivalent	Housing Equivalent	Macro Equivalent
Grouping Variable			
Sector (Retail, Office, Industrial)	Token Type (DeFi, CEX, NFT)	Region (Urban, Suburban, Rural)	Industry (Manufacturing, Services)
Outcome Variable			
Monthly Return (`ret`)	Token Return (`return_pct`)	Rent Growth (`rent_growth_yoy`)	Employment Growth (`emp_growth`)
Policy/Treatment Variable			

REIT Concept	Crypto Equivalent	Housing Equivalent	Macro Equivalent
Federal Funds Rate (`FEDFUNDS`)	Regulatory Sentiment (`reg_severity`)	Mortgage Rate (`MORTGAGE30US`)	Fiscal Shock (`govt_spending_chg`)
Control Variables			
Market Cap (`mcap`)	Trading Volume (`volume_usd`)	Median Home Price (`price_median`)	GDP (`gdp_real`)
Momentum Factor (`MOM`)	Price Momentum (`price_mom_6m`)	Price Appreciation (`price_mom_12m`)	Output Gap (`output_gap`)
Mechanism Channels			
Leverage costs	Liquidity disruption	Affordability shock	Credit tightening
Discount rate	Institutional exit	Supply response	Labor market adjustment
Demand effects (rental)	Network effects	Migration patterns	Consumption smoothing

1. Methodology

[Approximately 1 page. Explain data sources, sample construction, and model specifications **in plain language**.]

1.1 Data Sources

[Your Primary Dataset]

- **Source:** [Database name, provider, or instructor-provided file]
- **Coverage:** [N] unique [entities] ([entity type]: REITs, tokens, counties, firms), [frequency] observations from [start date] to [end date]
- **Variables:** [Outcome variable], [entity identifiers], [grouping variable if applicable], [time-varying characteristics]

REIT Example:

- Source: CRSP/Ziman Real Estate Database (instructor-provided)
- Coverage: 532 unique US equity REITs, monthly observations from January 2015 to December 2023
- Variables: Total returns (`ret`), market capitalization (`mcap`), sector classification, share price

Crypto Example:

- Source: CoinGecko API + Messari Crypto Data
- Coverage: 150 cryptocurrencies (tokens with >$50M market cap), daily observations from January 2020 to December 2024
- Variables: Daily returns (`return_pct`), market cap (`mcap_usd`), token type (DeFi/CEX/NFT), trading volume

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[Supplementary Data: Control Variables or Policy Indicators]

- **Source:** [Database name, API, or research paper]
- **Coverage/Frequency:** [Match primary dataset frequency or explain lag structure]
- **Variables:** [List 3-5 key variables with brief descriptions]

REIT Example:

- Source: Federal Reserve Economic Data (FRED) via API
- Series:
  - `FEDFUNDS`: Effective Federal Funds Rate (%)
  - `MORTGAGE30US`: 30-Year Fixed Mortgage Rate (%)
  - `CPIAUCSL`: Consumer Price Index (All Urban Consumers)
  - `UNRATE`: Unemployment Rate (%)

Crypto Example:

- Source: Regulatory Events Database (Custom-built from SEC press releases)
- Variables:
  - `reg_severity`: Regulatory announcement severity (scale 1-10, hand-coded)
  - `enforcement_count`: Monthly SEC enforcement actions against crypto firms
  - `exchange_delisting`: Binary indicator for exchange restrictions

[Optional: Factor Premiums or Risk Measures] *Only include if you constructed domain-specific factors (e.g., REIT factors, crypto liquidity metrics).*

REIT Example:

- Source: Seagraves et al. (2025) research data
- Factors: SIZE, VALUE, MOM (Momentum), QLTY (Quality), LOWVOL (Low Volatility), REV (Reversal)
- Description: Monthly factor returns based on tercile portfolio sorts

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1.2 Sample Construction

Panel Structure:

- **Initial sample:** ☑ [entities] × [Y] [time periods] = [Z] total observations
- **Entity type:** [REITs, tokens, counties, firms, countries]
- **Time unit:** [Monthly, daily, quarterly, annual]

Data Cleaning Steps:

- Removed duplicate [entity-time] pairs ([N] rows)
- Dropped observations with missing [outcome variable] ([N] rows)
- **[Dataset-specific cleaning]:** [Describe domain-appropriate filters]
  - *REIT example:* Winsorized extreme returns (>200% or <-100%) at 99th/1st percentiles; excluded small-cap REITs (market cap < $10M)

- *Crypto example:* Excluded stablecoins (low volatility by design); dropped tokens with <30 days trading history; capped returns at ±100% (flash crash protection)
- *Housing example:* Excluded counties with <50 transactions/month (noisy estimates); imputed missing rent data using regional medians

Final Sample: [Z] [entity-time] observations **Panel Type:** [Balanced / Unbalanced] ([explain why, e.g., entities enter/exit market])

1.3 Model Specifications

Model A: Fixed Effects Regression (Primary Specification)

We estimate the causal effect of [DRIVER] on [OUTCOME] using a two-way fixed effects panel regression:

```
[OUTCOME]_it = β₀ + β₁·[DRIVER]_lag_it + β₂·[CONTROL_1]_it + β₃·[CONTROL_2]_it +
α_i + δ_t + ε_it
```

Where:

- **[OUTCOME]_it:** [Your dependent variable description] for entity i in time t
- **[DRIVER]_lag_it:** [Your key treatment/policy variable] with [X-period] lag (based on M2 EDA lag analysis)
- **[CONTROL_1], [CONTROL_2]:** [Control variables that vary across entities and time]
- **α_i:** Entity fixed effect (controls for time-invariant entity characteristics)
- **δ_t:** Time fixed effect (controls for aggregate time shocks affecting all entities)
- **ε_it:** Error term (clustered standard errors at entity level)

Rationale: Fixed Effects controls for unobserved entity-specific factors that might confound the [DRIVER] effect. By comparing each entity to itself over time, we isolate the causal effect of [DRIVER] changes.

REIT Example:

```
ret_it = β₀ + β₁·FEDFUNDS_lag2_it + β₂·MOM_it + β₃·QLTY_it + β₄·SIZE_it + α_i +
δ_t + ε_it
```

- **ret_it:** Monthly return for REIT i in month t
- **FEDFUNDS_lag2_it:** Federal Funds Rate with 2-month lag
- **α_i:** Controls for time-invariant REIT characteristics (management quality, property portfolio)
- **Rationale:** Comparing each REIT to itself over time eliminates bias from unobserved REIT-specific factors

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Crypto Example:

```
return_it = β₀ + β₁·reg_severity_lag1_it + β₂·volume_it + β₃·btc_return_it + α_i +
δ_t + ε_it
```

- **return_it:** Daily return for token i on day t
- **reg_severity_lag1_it:** Regulatory severity index with 1-day lag
- **volume_it:** Log trading volume (liquidity control)
- **btc_return_it:** Bitcoin return (market factor)
- **α_i:** Controls for time-invariant token characteristics (blockchain, use case)
- **Rationale:** Comparing each token to itself over time eliminates bias from technology differences

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Model B: [Difference-in-Differences / ARIMA / ML Comparison]

[Describe your chosen Model B following the same structure as Model A. Include specification, variables, and rationale.]

When to use each Model B option:

- **Difference-in-Differences (DiD):** If you have a clear treatment/control group and a policy shock (e.g., Fed rate hike, regulatory change)
- **ARIMA:** If you have a single time series or want to forecast future values
- **ML Comparison (Random Forest, XGBoost):** If you want to compare linear model to non-linear alternatives

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DiD Example (REIT):

```
ret_it = β₀ + β₁·Sensitive_i + β₂·Post2022_t + β₃·(Sensitive_i × Post2022_t) +
controls + ε_it
```

- **Sensitive_i = 1** if REIT is in Retail or Office sector (rate-sensitive), 0 if Industrial (resilient)
- **Post2022_t = 1** if month ≥ January 2022 (rate hike period), 0 otherwise
- **β_3:** DiD estimator (differential effect on sensitive sectors post-shock)
- **Rationale:** Exploits natural experiment of Fed rate hike to identify causal effects. Assumes parallel trends (tested with placebo test).

DiD Example (Crypto):

```
return_it = β₀ + β₁·DeFi_i + β₂·PostSEC_t + β₃·(DeFi_i × PostSEC_t) + controls +
ε_it
```

- **DeFi_i = 1** if token is DeFi protocol, 0 if centralized exchange token
- **PostSEC_t = 1** if date ≥ June 2023 (SEC DeFi crackdown), 0 otherwise
- **β_3:** DiD estimator (differential effect on DeFi tokens post-regulation)

- **Rationale:** Compares DeFi tokens to CEX tokens (control group) before/after regulatory shock

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2. Results

[Approximately 1.5-2 pages. Present regression tables, figures, and interpretation.]

2.1 Fixed Effects Model (Main Specification)

Table 1: Fixed Effects Regression Results

Variable	Coefficient	Std. Error	t-statistic	p-value	Significance
[DRIVER] (lag X)	[β_1]	[SE]	[t]	[p]	[✓/✗]
[CONTROL_1]	[β_2]	[SE]	[t]	[p]	
[CONTROL_2]	[β_3]	[SE]	[t]	[p]	
Entity Fixed Effects	Yes				
Time Fixed Effects	Yes				
N (observations)	[N]				
R^2 (within)	[R^2]				

Notes: Clustered standard errors at [entity] level. *** p<0.01, ** p<0.05, * p<0.10.

REIT Example:

Variable	Coefficient	Std. Error	t-statistic	p-value	Significance
FEDFUNDS (lag 2)	-0.025	0.010	-2.50	0.012	**
MOM	0.042	0.013	3.23	0.001	***
QLTY	0.031	0.015	2.07	0.039	**
SIZE	-0.008	0.011	-0.73	0.465	
REIT Fixed Effects	Yes				
Time Fixed Effects	Yes				
N (observations)	54,320				
R^2 (within)	0.38				

Interpretation Framework

[Paragraph 1: Main coefficient interpretation with units]

The coefficient on [DRIVER] ([β_1], p = [p-value]) indicates that a **[1 unit] increase in [DRIVER] changes [OUTCOME] by [β_1 units]**, holding constant entity and time fixed effects.

Example interpretation template:

- ☑ percentage points"
- "A one-standard-deviation increase in [DRIVER] is associated with a [Y]% change in [OUTCOME]"

[Paragraph 2: Economic significance with context]

This effect is:

- **Economically significant:** [Provide real-world context. Example: "During the 2022-2023 rate hike cycle (0% → 5.25%), the model predicts a cumulative [X]% decline in [OUTCOME]"]
- **Statistically robust:** Survives multiple robustness checks ([list 2-3 checks from M3])

[Paragraph 3: Transmission mechanisms]

Consistent with economic theory:

- **Channel 1:** [Explain first causal mechanism]
- **Channel 2:** [Explain second causal mechanism]
- **[Optional] Channel 3:** [Explain third mechanism if applicable]

Use domain-appropriate mechanisms:

- **REIT mechanisms:** Leverage channel (debt financing costs), discount rate channel (DCF valuation), demand channel (rental market effects)
- **Crypto mechanisms:** Liquidity channel (exchange restrictions), institutional channel (fund flows), network effects (adoption)
- **Housing mechanisms:** Affordability channel (purchasing power), supply response (construction), regional variation (elasticity differences)

REIT Example Interpretation:

The coefficient on FEDFUNDS_lag2 (-0.025, p = 0.012) indicates that a **1 percentage point increase in the Federal Funds Rate reduces REIT returns by 2.5 percentage points**, holding constant REIT and time fixed effects. This effect is:

- **Economically significant:** During the 2022-2023 rate hike cycle (0% → 5.25%), the model predicts a cumulative 13.1% decline in REIT returns (2.5% × 5.25 pp).
- **Statistically robust:** Survives multiple robustness checks (alternative lags, exclusion of COVID crash, sector subsamples).
- **Consistent with economic theory:**
    - **Leverage channel:** REITs are highly leveraged (~60% debt-to-assets). Rising rates increase financing costs, reducing profitability.
    - **Discount rate channel:** REITs are valued based on discounted future cash flows. Higher rates increase the discount factor, lowering present value.

[Paragraph 4: Control variable interpretation]

Briefly interpret 1-2 control variables if they are economically meaningful. Skip if controls are purely technical.

REIT Example:

- **Momentum (MOM):** 1 SD increase in momentum premium → 4.2% higher REIT returns. Confirms trend-following behavior.
- **Quality (QLTY):** 1 SD increase in quality score → 3.1% higher returns. REITs with stable cash flows outperform.
- **Size (SIZE):** No significant effect after controlling for factors and fixed effects.

2.2 [Model B Results]

Table 2: [DiD / ARIMA / ML] Results

[Insert your Table 2 here, following the same format as Table 1]

Interpretation:

[Interpret Model B results following the same structure: coefficient estimates, economic significance, statistical robustness, consistency with theory]

2.3 Group Heterogeneity Analysis (If Applicable)

When to include this section:

- Your dataset has meaningful groups (sectors, asset types, regions, size quartiles)
- You expect differential sensitivity across groups (tested in M2 EDA)

If no natural grouping exists:

- Replace this section with **robustness checks** (split sample by time period, alternative variable definitions)
- Or skip and add discussion in Limitations section

Table 3: Sector-Specific Sensitivity (Optional)

Group	N	Correlation with [DRIVER]	Mean [OUTCOME]	Interpretation
[Group 1]	[N]	[ρ]	[μ]	[Most/Least sensitive]
[Group 2]	[N]	[ρ]	[μ]	
[Group 3]	[N]	[ρ]	[μ]	

REIT Example:

Sector	N	Correlation with FEDFUNDS	Mean Return	Interpretation

Sector	N	Correlation with FEDFUNDS	Mean Return	Interpretation
Retail	8,420	-0.45	-0.012	Highly sensitive (consumer spending)
Office	6,890	-0.42	-0.015	Highly sensitive (financing costs)
Industrial	5,120	-0.15	0.008	Resilient (e-commerce demand)

Crypto Example:

Token Type	N	Correlation with Reg Severity	Mean Return	Interpretation
DeFi	2,450	-0.58	-0.025	Highly sensitive (regulatory risk)
CEX	1,820	-0.32	-0.010	Moderate sensitivity
NFT	980	-0.21	-0.035	Low sensitivity (different risk factors)

2.4 Visual Evidence

Figure 1: [OUTCOME] vs. [DRIVER] (Dual-Axis Plot)

Dual-Axis Plot

Caption Template: *[OUTCOME] ([color], left axis) exhibits [positive/inverse/no] co-movement with [DRIVER] ([color], right axis). The [time period] [event description] coincides with [outcome behavior], consistent with our regression estimates.*

REIT Example:

Dual-Axis Plot

Caption: Aggregate REIT returns (blue, left axis) exhibit inverse co-movement with the Federal Funds Rate (red, right axis). The 2022-2023 rate hike cycle coincides with declining REIT performance, consistent with our regression estimates.

Crypto Example:

Dual-Axis Plot

Caption: Cryptocurrency returns (blue, left axis) exhibit sharp declines during regulatory crackdowns (red, right axis). The June 2023 SEC enforcement wave coincides with -15% average token returns, consistent with our DiD estimates.

Figure 2: [Group] Sensitivity to [DRIVER] (If Applicable)

Sensitivity Bar Chart

Caption Template: *[Group A] and [Group B] are highly sensitive to [DRIVER] (correlations: [ρ_1], [ρ_2]), while [Group C] is resilient ([ρ_3]). This heterogeneity informs our [audience-specific] recommendations.*

REIT Example:

Sector Sensitivity Bar Chart

Caption: Retail and Office REITs are highly sensitive to interest rate changes (correlations: -0.45, -0.42), while Industrial REITs are resilient (-0.15). This heterogeneity informs our sector-specific investment recommendations.

Figure 3: Residuals vs. Fitted Values (Diagnostic Plot)

Residuals Plot

Caption (Generic): *Residuals scatter randomly around zero with no clear pattern, suggesting homoskedasticity (after applying clustered standard errors). Q-Q plot (not shown) indicates approximate normality with [fat/thin] tails, which is [typical/atypical] for [data type].*

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3. Conclusions & Recommendations

[Approximately 1 page. Translate findings into actionable advice for your decision-maker audience.]

3.1 [Audience-Specific Recommendations]

Choose the recommendation framing appropriate for your audience:

Option 1: Investment Committee (REITs, Stocks, Crypto, Assets)

Based on our empirical analysis, we provide the following **portfolio recommendations** for the current [market condition]:

Recommended [Sector/Asset/Group] Allocation (Tactical):

1. Overweight [Group A] ([X]% allocation above benchmark)

- **Rationale:** [Group A] exhibits [low/high] sensitivity to [DRIVER] (correlation = [ρ]). [Domain-specific fundamentals explanation].
- **Expected performance:** [Outperformance/hedging] relative to other groups in [current environment].

2. Neutral-to-Underweight [Group B]

- **Rationale:** [Group B] is highly sensitive to [DRIVER] (correlation = [ρ]). Expected to [underperform/outperform] until [condition changes].
- **Caveat:** Selective exposure to high-quality [subset] may offer value.

REIT Investment Committee Example:

Based on our empirical analysis, we provide the following **portfolio recommendations** for the current interest rate environment (Federal Funds Rate: 5.0-5.25%):

Recommended Sector Allocation (Tactical):

1. Overweight Industrial REITs (15% allocation above benchmark)

- **Rationale:** Industrial REITs exhibit low interest rate sensitivity (correlation = -0.15) and are driven by structural demand from e-commerce and logistics. Sector fundamentals (supply constraints, long-term leases) are inelastic to monetary policy.
- **Expected performance:** Outperformance relative to other REIT sectors in elevated rate environment.

2. Neutral-to-Underweight Retail and Office REITs

- **Rationale:** These sectors are highly rate-sensitive (correlations: -0.45, -0.42) due to consumer spending sensitivity and high leverage. Expected to underperform until rates stabilize or decline.
- **Caveat:** Selective exposure to high-quality Retail REITs (e.g., prime shopping centers with anchor tenants) may offer value, but broad sector exposure is not recommended.

Crypto Portfolio Example:

Based on our empirical analysis, we provide the following **portfolio recommendations** for the current regulatory environment (ongoing SEC enforcement):

Recommended Token Type Allocation (Tactical):

1. Overweight Bitcoin/Ethereum (30% allocation above benchmark)

- **Rationale:** BTC and ETH have established regulatory frameworks (CFTC commodity classification, ETF approvals). Low sensitivity to SEC enforcement actions.
- **Expected performance:** Defensive positioning in uncertain regulatory environment.

2. Reduce DeFi Token Exposure (-20% relative to benchmark)

- **Rationale:** DeFi tokens are highly sensitive to regulatory crackdowns (correlation = -0.58). Ongoing SEC scrutiny of DeFi protocols creates tail risk.
- **Caveat:** If Congress passes comprehensive crypto legislation (30% probability over 12 months), DeFi could recover 15-20%.

Option 2: Policy Committee (Economic Indicators, Labor Data)

Based on our empirical analysis, we provide the following **policy recommendations** for [government agency / central bank]:

Policy Implication 1:

- **Finding:** [Summary of main effect]
- **Recommendation:** [Policy action to address finding]

- **Rationale:** [Economic mechanism + distributional effects]

Policy Implication 2:

- **Finding:** [Group heterogeneity result]
- **Recommendation:** [Targeted intervention for vulnerable groups]
- **Rationale:** [Equity considerations]

Housing Policy Example:

Based on our empirical analysis, we provide the following **policy recommendations** for the Federal Reserve:

Policy Implication 1: Interest Rate Sensitivity Varies by Region

- **Finding:** A 1 percentage point increase in mortgage rates reduces home sales by 12% in supply-constrained markets (San Francisco, New York) but only 4% in supply-elastic markets (Houston, Dallas).
- **Recommendation:** Consider regional heterogeneity when setting monetary policy. Rate hikes have disproportionate effects on coastal housing markets.
- **Rationale:** Supply elasticity determines price vs. quantity adjustment. Inelastic markets experience larger price declines and affordability shocks.

Policy Implication 2: First-Time Buyers Disproportionately Affected

- **Finding:** Rate increases reduce first-time buyer share by 8 percentage points, while repeat buyers (less leverage-dependent) are unaffected.
- **Recommendation:** Pair rate hikes with first-time buyer assistance programs (down payment grants, FHA loan expansion) to mitigate distributional effects.

Option 3: Risk Committee (Banking, Insurance, Internal Risk)

Based on our empirical analysis, we provide the following **risk management recommendations**:

Risk Exposure 1:

- **Finding:** [Quantify risk exposure to DRIVER]
- **Recommendation:** [Hedging strategy or exposure limit]
- **Implementation:** [Specific actions]

Risk Exposure 2:

- **Finding:** [Tail risk or non-linear effect]
- **Recommendation:** [Stress testing or scenario planning]

3.2 [Scenario Analysis]

We model [2-4] scenarios for the next [time period] based on [event/policy expectations]:

Scenario	[DRIVER] Change	Predicted [OUTCOME] Impact	Probability

Scenario	[DRIVER] Change	Predicted [OUTCOME] Impact	Probability
Baseline: [Description]	[X]	[Impact]	[%]
Optimistic: [Description]	[Y]	[Impact]	[%]
Pessimistic: [Description]	[Z]	[Impact]	[%]

Expected value: Weighted average [outcome] = [calculation]

Recommendation: Given [asymmetric upside/downside], a [stance description] is warranted.

REIT Example:

Scenario	Rate Change	Predicted REIT Return Impact	Probability (Futures Markets)
Baseline: Rates hold at 5.0-5.25%	0 bp	Flat returns (0% to +2%)	40%
Dovish: Fed cuts 100 bp to 4.0-4.25%	-100 bp	+2.5% recovery (with 2-month lag)	35%
Hawkish: Fed raises 50 bp to 5.5-5.75%	+50 bp	-1.25% decline	25%

Expected value: Weighted average return = (0.40 × 1%) + (0.35 × 2.5%) + (0.25 × -1.25%) = **+0.96%**

Recommendation: Given asymmetric upside (dovish scenario) and limited downside (rate hikes unlikely beyond 50 bp), a **neutral-to-slight-overweight** stance on aggregate REIT exposure is warranted.

3.3 Risk Assessment

Model Risks:

1. **Assumption of stable elasticity:** Our model assumes the historical relationship ([β] = [value]) will hold in the future. If structural changes have altered this relationship, predictions may be biased.
2. **Omitted variable bias:** [List 2-3 potentially confounding factors]. If these correlate with [DRIVER], our coefficient estimates may be confounded.
3. **External validity:** Results are based on [time period] data. May not generalize to [different context].

[Domain-Specific] Risks:

1. **[Risk 1]:** [Describe risk and potential impact]
2. **[Risk 2]:** [Describe risk and potential impact]
3. **[Risk 3]:** [Describe risk and potential impact]

REIT Example:

Market Risks:

1. **Rate volatility:** If the Fed reverses course (multiple rate cuts followed by hikes), REIT returns will exhibit increased volatility, making forecasts less reliable.
2. **Sector-specific shocks:** Retail REITs face structural headwinds (e-commerce disruption) independent of rates. Our sector recommendations assume no exogenous shocks.
3. **Liquidity risk:** In a market downturn, REIT liquidity may decline, amplifying volatility and reducing the effectiveness of tactical rebalancing.

3.4 Caveats and Limitations

Model Limitations:

1. **Fixed Effects assumption:** We assume [entity] characteristics are time-invariant. If [entities] change structure during sample period, our FE estimates may be biased.
2. **[Model B] assumption:** [State key assumption for Model B, e.g., parallel trends for DiD, stationarity for ARIMA]. [Evidence for/against assumption].
3. **Lag specification:** We chose a [X-period] lag based on M2 EDA. Alternative lag structures yield similar but not identical results. The true lag may vary by [group/context].
4. **Measurement error:** [Describe any measurement challenges in your data]. Alternative definitions could yield different results.

REIT Example:

Caveats and Limitations:

1. **Fixed Effects assumption:** We assume REIT characteristics (management quality, property portfolio) are time-invariant. If REITs restructure portfolios during the sample period (e.g., exit retail, enter industrial), our FE estimates may be biased.
2. **Parallel trends (DiD):** Our DiD analysis assumes sensitive and resilient sectors would have followed parallel trends absent the rate shock. Pre-trend divergence in 2019-2020 suggests this assumption may be violated, potentially overstating the causal effect.
3. **Lag specification:** We chose a 2-month lag based on M2 EDA. Alternative lag structures (1-month, 3-month) yield similar but not identical results. The true lag may vary by sector or REIT size.
4. **Measurement error:** REIT factor premiums are based on tercile sorts, which may not fully capture factor exposure heterogeneity. Alternative factor construction methods (e.g., value-weighted, decile sorts) could yield different results.

3.5 Future Research Directions

To refine this analysis, future work could:

- **[Research direction 1]:** [Explain what additional data or methods would improve the analysis]
- **[Research direction 2]:** [Explain how to test key assumptions]
- **[Research direction 3]:** [Explain how to extend external validity]

REIT Example:

To refine this analysis, future work could:

- **Incorporate investor sentiment data** (VIX, REIT fund flows) to control for market psychology
- **Analyze heterogeneous treatment effects** by REIT size, leverage, or geographic concentration
- **Extend the sample period** to include earlier rate cycles (1990s, 2000s) to test external validity
- **Model non-linearities** (e.g., threshold effects where rate increases above 5% have disproportionate impacts)

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4. References

[List all data sources, academic papers (if cited), and external references in APA or similar format]

Required:

1. **[Your primary data source]** (Year). *[Dataset name and description].* Retrieved from [URL or "Provided by instructor"]
2. **[Supplementary data source]** (Year). *[Dataset name].* Retrieved from [URL]

Optional (if cited): 3. **[Academic paper from Orbis Lit-Anchor]:** Author, A., & Author, B. (Year). Title. *Journal Name, Volume*(Issue), pages.

REIT Example:

1. **Federal Reserve Economic Data (FRED).** (2024). *Effective Federal Funds Rate, 30-Year Mortgage Rate, CPI, Unemployment Rate.* Retrieved from https://fred.stlouisfed.org
2. **CRSP/Ziman Real Estate Database.** (2024). *REIT Master Panel: Monthly returns and characteristics for US equity REITs.* Provided by instructor.
3. **Letdin, M., et al.** (2026). *REIT Factors.* Working paper.
4. **Smith, J., & Johnson, L.** (2020). Interest rate sensitivity of real estate investment trusts. *Journal of Financial Economics, 125*(3), 450-472.

Appendix: AI Audit Summary

[0.5-1 page. Summarize AI use across all milestones. **MANDATORY - Missing = 0/50 points**]

AI Tools Used

- [List all AI tools: ChatGPT, Claude, GitHub Copilot, etc.]
- [Specify model versions if known: GPT-4, Claude Sonnet, etc.]

Key Verification Examples

M1 Example:

- **Prompt:** "[Your actual prompt to AI]"
- **Output:** [Brief summary or code snippet AI provided]
- **Verification:** [How you tested/validated the output]
- **Critique:** [What you changed or what AI got wrong]

M2 Example:

- **Prompt:** "[Your actual prompt to AI]"
- **Output:** [Brief summary or code snippet AI provided]
- **Verification:** [How you tested/validated the output]
- **Critique:** [What you changed or what AI got wrong]

M3 Example:

- **Prompt:** "[Your actual prompt to AI]"
- **Output:** [Brief summary or code snippet AI provided]
- **Verification:** [How you tested/validated the output]
- **Critique:** [What you changed or what AI got wrong]

M4 Example:

- **Prompt:** "[Your actual prompt to AI for memo writing/editing]"
- **Output:** [Brief summary of AI response]
- **Verification:** [How you fact-checked interpretation/recommendations]
- **Critique:** [What domain knowledge did you add that AI missed?]

REIT Example (M3):

- **Prompt (ChatGPT):** "Interpret a Fixed Effects coefficient of -0.025 for FEDFUNDS."
- **Output:** "A 1% increase in FEDFUNDS reduces returns by 2.5%."
- **Critique:** Phrasing is ambiguous ("1%" vs. "1 pp"). Corrected to: "A 1 percentage point increase in FEDFUNDS reduces returns by 2.5 percentage points."

Responsibility Statement

All code and analysis in this memo has been verified by our team. We used AI as a **productivity tool**, not as a substitute for understanding. We take full responsibility for any errors and do not claim "the AI did it" as an excuse.

END OF MEMO

Team Members:

- [Name 1]
- [Name 2]
- [Name 3]
- [Name 4] (if applicable)

Submission Date: [Date]

Course: QM 2023: Statistics II / Data Analytics, Spring 2026, University of Tulsa